

Linear Programming through Excel Solver

Before moving on to excel solver, we need to have detailed question. Therefore, we start with the following problem:

A firm makes two types of furniture, chairs and tables. Profits are ₹ 20 per chair and ₹ 30 per table. Both products are processed on three machines M1, M2 and M3. The time required for each product in hours and total time available in hours per week on each machine are as follows:

Machine	Chair	Table	Available Time
M1	3	3	36
M2	5	2	50
M3	2	6	60

How should the manufacturer schedule his production to maximize profit?

STEP 1: Formulate the problem:

Let x and y be the number of chairs and tables produced per week respectively.

The LPP model is:

Maximize $Z = 20x + 30y$

Subject to constraints:

$$3x + 3y \leq 36$$

$$5x + 2y \leq 50$$

$$2x + 6y \leq 60$$

$$x, y \geq 0 \text{ (non-negativity constraint)}$$

STEP 2: Now enter all this information to excel spreadsheet in the below format.

	A	B	C	D	E	F	G
1		Chair	Table	Profit			
2	Decision variables						
3	Contribution	20	30				
4							
5	Constraints				Maximum Capacity		
6	Machine 1	3	3		36		
7	Machine 2	5	2		50		
8	Machine 3	2	6		60		
9							
10	NOTE: In the shaded cells we need to apply the formulas						
11							
12							

STEP 3: NEXT, enter the formula in cell D3 and D6 to D8.

	A	B	C	D	E	F	G	H
1		Chair	Table	Profit				
2	Decision variables							
3	Contribution	20	30	=SUMPRODUCT(\$B\$2:\$C\$2,B3:C3)				
4				SUMPRODUCT(array1, [array2], [array3], [array4], ...)				
5	Constraints				Maximum Capacity			
6	Machine 1	3	3		36			
7	Machine 2	5	2		50			
8	Machine 3	2	6		60			
9								
10	NOTE: In the shaded cells we need to apply the formulas							
11								

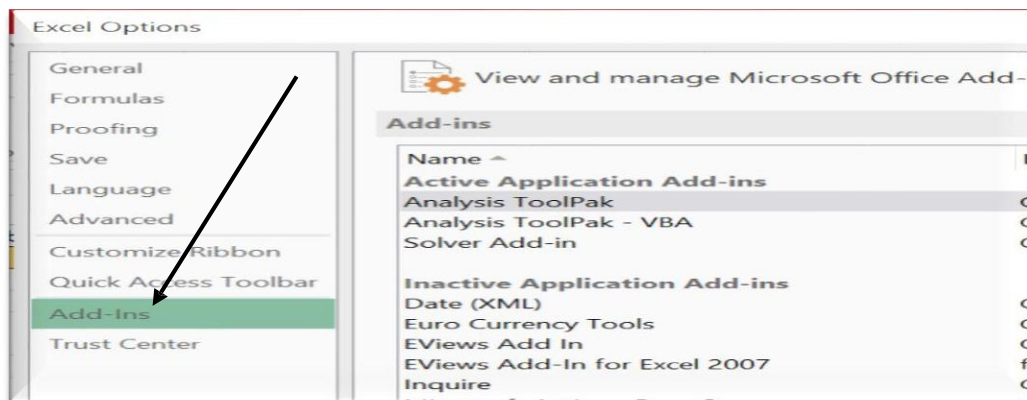
Enter the formula in D3 to calculate total profit using SUMPRODUCT formula (i.e. multiplying profit per unit into quantity of specific furniture and then adding it)

Similarly enter the formula in D6 to D8 using SUMPRODUCT formula to calculate capacity utilized of each machine

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW								
Cut Copy Paste Format Painter Font Font Alignment Alignment								
D6 : X ✓ fx =SUMPRODUCT(\$B\$2:\$C\$2,B6:C6)								
	A	B	C	D	E	F	G	H
1		Chair	Table	Profit				
2	Decision variables							
3	Contribution	20	30	0				
4								
5	Constraints				Maximum Capacity			
6	Machine 1	3	3	=SUMPRODUCT(\$B\$2:\$C\$2,B6:C6)				
7	Machine 2	5	2		50			
8	Machine 3	2	6		60			
9								
10	NOTE: In the shaded cells we need to apply the formulas							
11								

STEP 4: To load the solver add-in, execute the following steps.

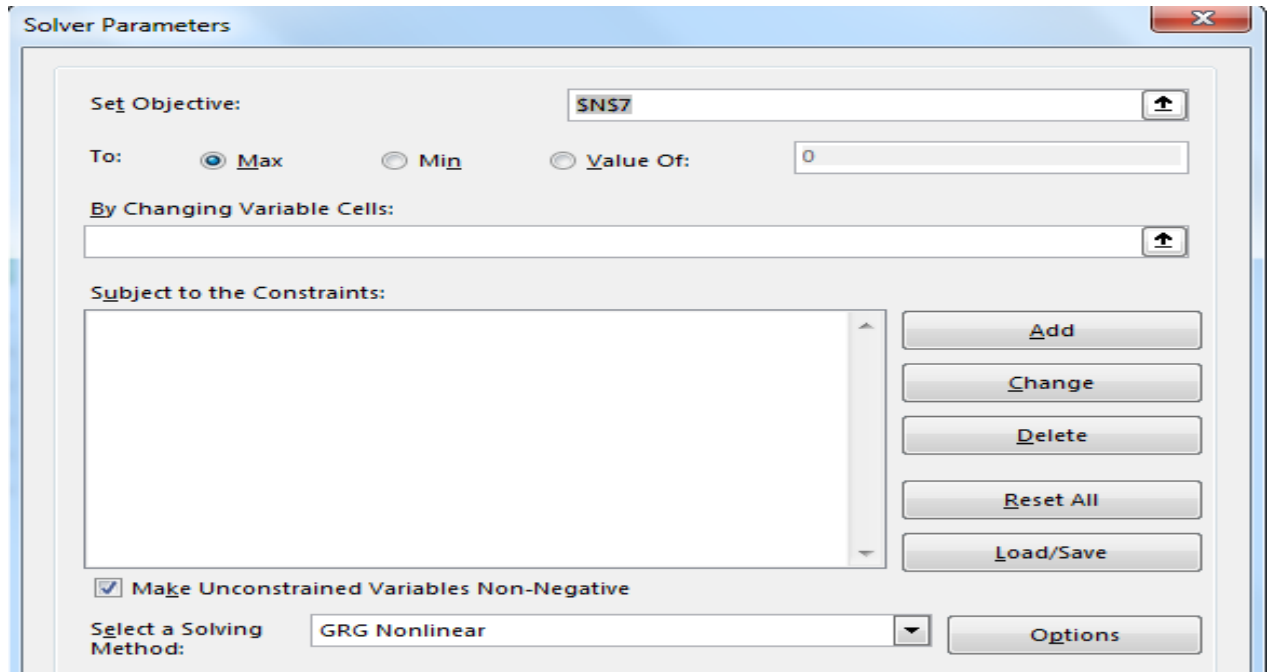
- Go to the File tab then, under File tab click on “Options”.
- Go to Add-ins under Excel Options.



- Under this select “Excel Add-ins” and click on Go.

STEP 5: Solve Linear Programming through Excel Solver

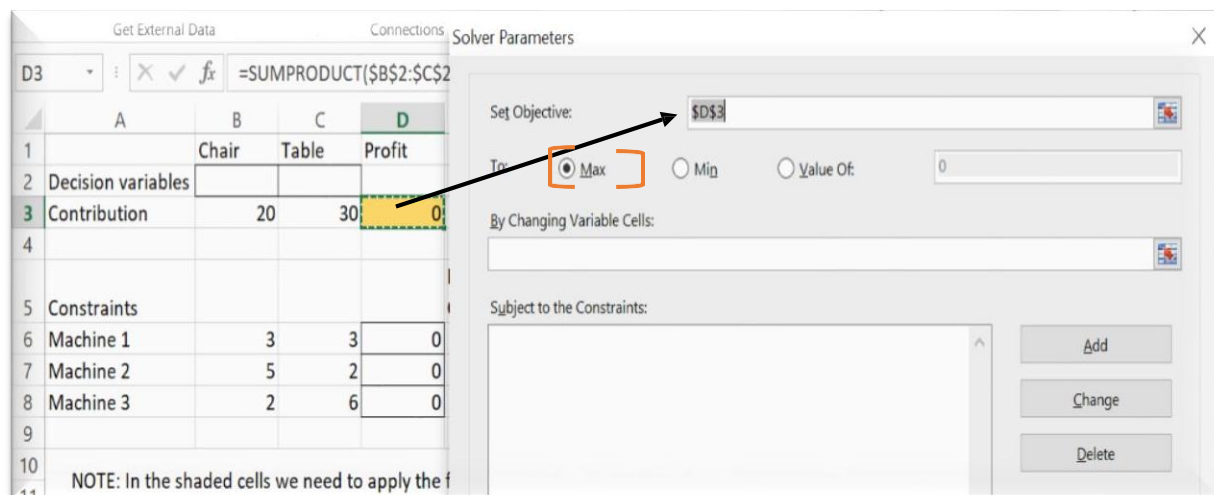
To apply solver go to the DATA tab and click on “Solver” we will see below window.



The Solver Parameters window is shown with the following settings:

- Set Objective:** \$N\$7
- To:** ☒ Max ☐ Min ☐ Value Of: 0
- By Changing Variable Cells:** (empty)
- Subject to the Constraints:** (empty)
- ☒ Make Unconstrained Variables Non-Negative
- Select a Solving Method:** GRG Nonlinear
- Buttons:** Add, Change, Delete, Reset All, Load/Save, Options

- In the above window, our first option is “Set Objective”. Our objective is to identify the “Total Profit”, so our Total Profit cell is D3, so select the cell D3 for this “Set Objective” and set it to “Max”.



The Solver Parameters window is overlaid on an Excel spreadsheet. The spreadsheet data is as follows:

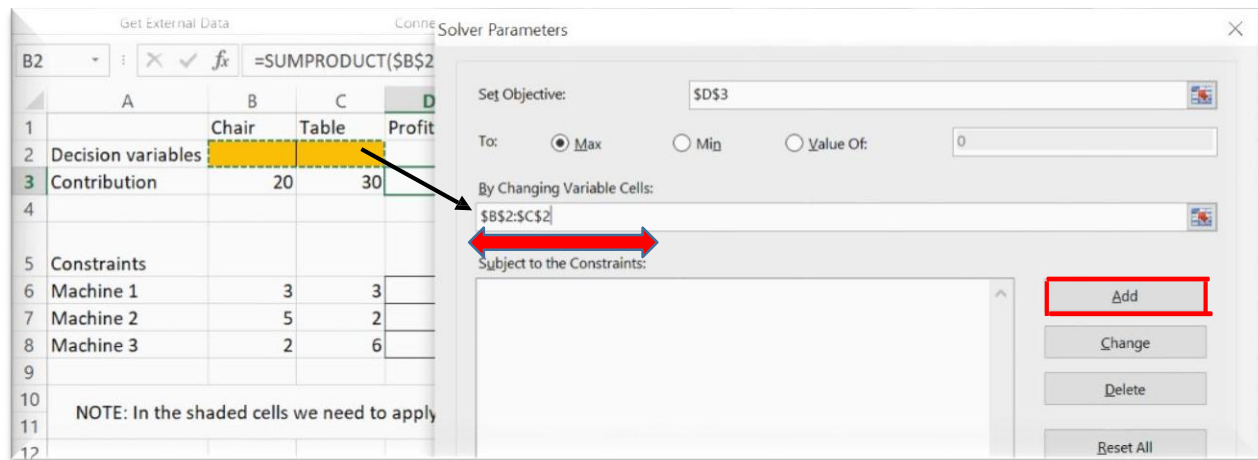
	A	B	C	D
1		Chair	Table	Profit
2	Decision variables			
3	Contribution	20	30	0
4				
5	Constraints			
6	Machine 1	3	3	0
7	Machine 2	5	2	0
8	Machine 3	2	6	0
9				
10				
11				

The Solver Parameters window is configured as follows:

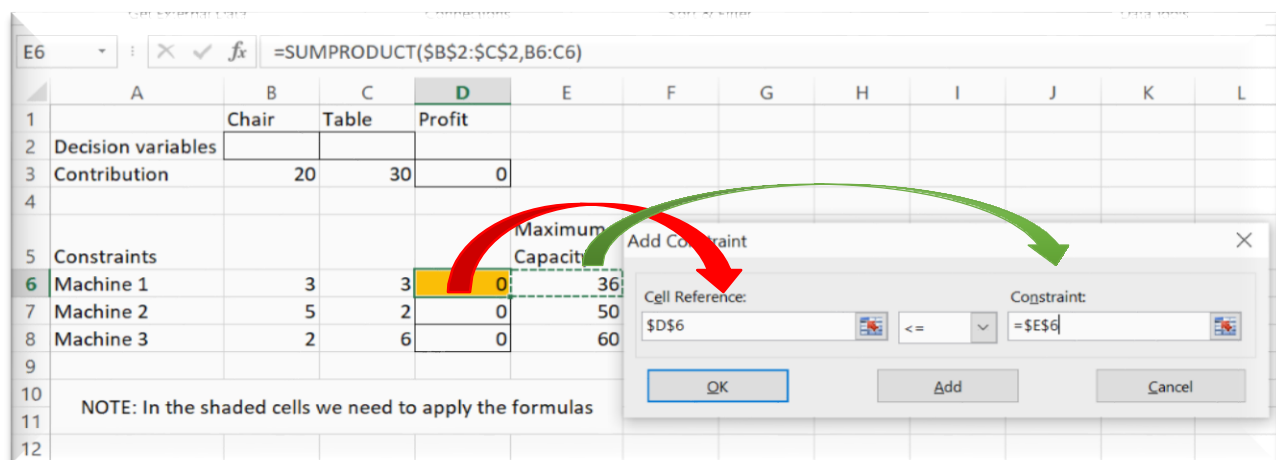
- Set Objective:** \$D\$3 (indicated by an arrow from cell D3 in the spreadsheet)
- To:** ☒ Max ☐ Min ☐ Value Of: 0
- By Changing Variable Cells:** (empty)
- Subject to the Constraints:** (empty)
- Buttons:** Add, Change, Delete

NOTE: In the shaded cells we need to apply the f

- Next option is “By changing variables”, in this problem, our variables are “Chairs” and “Tables”. To select a range of cell B2:C2.



- Now, the next option “subject to the constraints”, Once you click on “Add” we will see below add constraint window.



- Click on “Add” to stay back in the same window. Now in the second constraint select the value as D7 and select “<=” and under constraint select G7 cell.
Similarly, for the third constraint select the value as D8 and select “<=” and under constraint select G8 cell.

- After adding all constraints, Click on “Ok” to come out of Add Constraint window. Constraints are added as below. Now all our parameters are ready.

The Solver Parameters dialog box is shown with the following settings:

- Set Objective:** \$D\$3
- To:** ☒ Max ☐ Min ☐ Value Of: 0
- By Changing Variable Cells:** \$B\$2:\$C\$2
- Subject to the Constraints:**
 - \$D\$6 <= \$E\$6
 - \$D\$7 <= \$E\$7
 - \$D\$8 <= \$E\$8
- Buttons:** Add, Change

- Now tick check box for **non-negative constraints** and select “**Simplex LP**” as solving method from the drop down. Finally, Click on **Solve**.

The Solver Parameters dialog box is shown with the following settings:

- Set Objective:** \$D\$3
- To:** ☒ Max ☐ Min ☐ Value Of: 0
- By Changing Variable Cells:** \$B\$2:\$C\$2
- Subject to the Constraints:**
 - \$D\$6 <= \$E\$6
 - \$D\$7 <= \$E\$7
 - \$D\$8 <= \$E\$8
- Buttons:** Add, Change, Delete, Reset All, Load/Save
- Make Unconstrained Variables Non-Negative:** ☒ (highlighted with a red box)
- Select a Solving Method:** Simplex LP (highlighted with a red box)
- Solving Method:** Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.
- Buttons:** Help, Solve (highlighted with a red box and a red arrow), Close

- Finally, the answers will appear on the screen as you can see in the shaded cells below.

	A	B	C	D	E
1		Chair	Table	Profit	
2	Decision variables	3	9		
3	Contribution	20	30	330	
4					
5	Constraints				Maximum Capacity
6	Machine 1	3	3	36	36
7	Machine 2	5	2	33	50
8	Machine 3	2	6	60	60
9					
10	NOTE: In the shaded cells we need to apply the formulas				
11					
12					
13					
14					
15					
16					

Solver Results

Solver found a solution. All Constraints and optimality conditions are satisfied.

☒ Keep Solver Solution
☐ Restore Original Values

☐ Return to Solver Parameters Dialog

☐ Outline Reports

Solver found a solution. All Constraints and optimality conditions are satisfied.

When the GRG engine is used, Solver has found at least a local optimal solution. When Simplex LP is used, this means Solver has found a global optimal solution.

Click Ok, to save the solution.

STEP 5: Write the final solution.

As per the solution given, firm can have maximum profit of ₹ 330 by producing 3 chairs and 9 tables per week.

End....

3.3 Linear Programming Applications

The areas covered by these applications include the following:

1. Investment
2. Production Planning and inventory control
3. Manpower planning
4. Urban development planning
5. Oil refining and blending.

3.3.1 Investment

Multitudes of investment opportunities are available to today's investor. Examples of investment problems are capital budgeting for projects, bond investment strategy, stock portfolio selection, and establishment of bank loan policy. In many of these situations, LP can be used to select the optimal mix of opportunities that will maximize return while meeting requirements set by the investor and the market.

Example 3.3.1. *Bank Loan Model*

Bank One is in the process of devising a loan policy that involves a maximum of \$ 12 million. The following table provides the pertinent data about available loans.

Type of loan	Interest rate	Bad-debt ratio
Personal	0.140	0.10
Car	0.130	0.07
Home	0.120	0.03
Farm	0.125	0.05
Commercial	0.100	0.02

Bad debts are unrecoverable and produce no interest revenue. Competition with other financial institutions dictates the allocation of at least 40% of the funds to farm and commercial loans. To assist the housing industry in the region, home loans must equal at least 50% of the personal, car and home loans. The bank limits the overall ratio of bad debts on all loans to at most 4%.

Mathematical Model: The situation deals with determining the amount of loan in each category, thus leading to the following definitions of the variables:

- x_1 = Personal loans (in millions of dollars)
- x_2 = car loans
- x_3 = home loans
- x_4 = farm loans
- x_5 = commercial loans

The objective of the Bank One is to maximize net return, the difference between interest revenue and lost bad debts. Interest revenue is accrued on loans in good standing. For example, when 10% of personal loans are lost to bad debt, the bank will receive interest on 90% of the loan—that is, it will receive 14% interest on $0.9x_1$ of the original loan x_1 . The same reasoning is applies to the remaining four other types of loans. Thus,

$$\begin{aligned}\text{Total interest} &= 0.14(0.90x_1) + 0.13(0.93x_2) + 0.12(0.97x_3) + 0.125(0.95x_4) + 0.10(0.98x_5) \\ &= 0.126x_1 + 0.1209x_2 + 0.1164x_3 + 0.11875x_4 + 0.098x_5\end{aligned}$$

We also have

$$\text{Bad debt} = 0.10x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4 + 0.02x_5$$

The objective function combines interest revenue and bad debt as

$$\begin{aligned}\text{Maximize } z &= \text{Total interest} - \text{Bad debt} \\ \text{Maximize } z &= (0.126x_1 + 0.1209x_2 + 0.1164x_3 + 0.11875x_4 + 0.098x_5) \\ &\quad - (0.10x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4 + 0.02x_5) \\ &= 0.026x_1 + 0.0509x_2 + 0.0864x_3 + 0.06875x_4 + 0.078x_5\end{aligned}$$

The problem has five constraints:

1. Total funds should not exceed \$12 (million):

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq 12$$

2. Farm and commercial loans equal at least 40% of all loans:

$$x_4 + x_5 \geq 0.4(x_1 + x_2 + x_3 + x_4 + x_5)$$

or

$$0.4x_1 + 0.4x_2 + 0.4x_3 - 0.6x_4 - 0.6x_5 \leq 0$$

3. Home loans should equal at least 50% of persoanl, car and home loans:

$$x_3 \geq 0.5(x_1 + x_2 + x_3)$$

or

$$0.5x_1 + 0.5x_2 - 0.5x_3 \leq 0$$

4. Bad debts should not exceed 4% of all loans:

$$0.10x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4 + 0.02x_5 \leq 0.04(x_1 + x_2 + x_3 + x_4 + x_5)$$

or

$$0.06x_1 + 0.03x_2 - 0.01x_3 + 0.01x_4 - 0.02x_5 \leq 0$$

5. Nonnegativity:

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0$$

A subtle assumption in the preceding formulation is that all loans are issued at approximately the same time. It allows us to ignore differences in the time value of the funds allocated to the different loans.

Solution: The optimal solution computed using Excel Solver yields:

M5										
	A	B	C	D	E	F	G	H	I	J
1	Bank One Problem									
2										
3		x_1: Personal loan, x_2: Car loan, x_3: Home loan, x_4: Farm loan, x_5: Commercial loan								
4										Net.return
5	Variables	x_1	x_2	x_3	x_4	x_5				Z
6	Solution	0	0	7.2	0	4.8				0.99648
7										
8	Objec.fun	0.026	0.0509	0.0864	0.06875	0.078				
9										
10										
11	Constraints							LHS		RHS
12	Constraint_1	1	1	1	1	1		12 <=		12
13	Constraint_2	0.4	0.4	0.4	-0.6	-0.6		-8.9E-16 <=		0
14	Constraint_3	0.5	0.5	-0.5				-3.6 <=		0
15	Constraint_4	0.06	0.03	-0.01	0.01	-0.02		-0.168 <=		0

Fig. 3: Results

That is,

$$z = 0.99648, x_1 = 0, x_2 = 0, x_3 = 7.2, x_4 = 0, x_5 = 4.8$$

Remarks: The optimal solution calls for allocating all \$12 million: \$7.2 million to home loans and \$4.8 million to commercial loans, to achieve a maximum net return of \$996,480. The remaining categories receive none.

The return on the investment is

$$\text{Rate of return} = \frac{z}{12} = \frac{0.99648}{12} = 0.08304$$

This shows that the combined annual rate of return is 8.034%.

3.3.2 Production Planning and Inventory Control

There is a wealth of LP applications to production and inventory control. Here is an example with production scheduling to meet a single-period of demand.

Example 3.3.2. (Single-period Production Model)

In preparation for winter season, a clothing company is manufacturing parka and goose overcoats, insulated pants, and gloves. All products are manufactured in four different departments: cutting, insulating, sewing, and packaging. The company has received firm orders for its products. The contract stipulates a penalty of undelivered items. Devise an optimal production plan for the company based on the following data:

Department	Time per unit (hr)				Capacity (hr)
	Parka	Goose	Pants	Gloves	
Cutting	.30	.30	.25	.15	1000
Insulating	.25	.35	.30	.10	1000
Sewing	.45	.50	.40	.22	1000
Packaging	.15	.15	.1	.05	1000
Demand	800	750	600	500	
Unit profit	\$30	\$40	\$20	\$10	
Unit penalty	\$15	\$20	\$10	\$8	

Mathematical Model: The variable of the problem are

x_1 = number of parka jackets

x_2 = number of goose jackets

x_3 = number of pairs of pants

x_4 = number of pairs of gloves

The company is penalised for not meeting demand. The objective then is to maximize the net profit, defined as

$$\text{Net profit} = \text{Total profit} - \text{Total penalty}$$

The total profit is $30x_1 + 40x_2 + 20x_3 + 10x_4$. To compute the total penalty, the demand constraints can be written as

$$x_1 + s_1 = 800, x_2 + s_2 = 750, x_3 + s_3 = 600, x_4 + s_4 = 500,$$

$$x_j \geq 0, s_j \geq 0, j = 1, 2, 3, 4$$

The new variable s_j represent the shortage in demand for product j , and the total penalty can be computed as $15s_1 + 20s_2 + 10s_3 + 8s_4$. The complete model thus becomes

$$\text{Maximize } z = 30x_1 + 40x_2 + 20x_3 + 10x_4 - (15s_1 + 20s_2 + 10s_3 + 8s_4)$$

Subject to

$$\begin{aligned}
.30x_1 + .30x_2 + .25x_3 + .15x_4 &\leq 1000 \\
.25x_1 + .35x_2 + .30x_3 + .10x_4 &\leq 1000 \\
.45x_1 + .50x_2 + .40x_3 + .22x_4 &\leq 1000 \\
.15x_1 + .15x_2 + .10x_3 + .05x_4 &\leq 1000
\end{aligned}$$

$$x_1 + s_1 = 800, x_2 + s_2 = 750, x_3 + s_3 = 600, x_4 + s_4 = 500$$

$$x_j \geq 0, s_j \geq 0, j = 1, 2, 3, 4$$

Solution:

The optimum solution (obtained using TORA) is $z = \$64,625$, $x_1 = 800$, $x_2 = 750$, $x_3 = 387.5$, $x_4 = 500$, $s_1 = s_2 = s_3 = 0$, $s_4 = 212.5$. The solution satisfies all the demand for both types of jackets and the gloves. A shortage of 213 (rounded up from 212.5) pairs of pants will result in a penalty cost of $213 \times \$10 = \2130 .

3.3.3 Urban development Planning

Urban planning deals with three general areas: (1) building new housing developments, (2) upgrading inner-city deteriorating housing and recreational areas, and (3) planning public facilities (such as schools and airports). The constraints associated with these projects are both economic (land, construction and financing) and social (schools, parks, and income level). The objectives in urban planning vary. In new housing developments, profit is usually the motive for undertaking the project. In the remaining two categories, the goals involve social, political, economic and cultural considerations.

Question: (Urban Renewal Model)

The city of Erstville is faced with a severe budget shortage. Seeking a long term solution, the city council votes to improve the tax base by condemning an inner-city housing area and replacing it with a modern development. The project involves two phases: (1) demolishing substandard houses to provide land of the new development and (2) building the new development. The following is a summary of the situation.

1. As many as 300 substandard houses can be demolished. Each house occupies a 0.25-acre lot. The cost of demolishing a condemned house is \$2000.
2. Lot sizes for new single-,double-,triple-, and quadruple-family homes (units) are 0.18, 0.28, 0.4, and 0.5 acre, respectively. Streets, open space, and utility elements account for 15% of available acreage.
3. In the new development, the tripple and quadruple units account for at least 25% of the total. Single units must be at least 20% of all units and double units at least 10%.
4. The Tax levied per unit for single, double, triple and quadruple units \$1000, \$1900, \$2700, and \$3400, respectively.

5. The construction cost per unit for single-, double-, triple-, and quadruple-family homes is \$50,000, \$70,000, \$130,000, and \$160,000, respectively. Financing through a local bank is limited to \$15 million.

How many units of each type should be constructed to maximize tax collection?

Solution:

Mathematical Model:

Besides determining the number of units to be constructed of each type of housing, we also need to decide how many houses must be demolished to make room of new development. Thus, the variables of the problem can be defined as follows:

x_1 = number of units of single-family homes

x_2 = number of units of double-family homes

x_3 = number of units of triple-family homes

x_4 = number of units of quadruple-family homes

x_5 = number of old homes to be demolished

The objective is to maximize total tax collection from all four types of homes-that is,

$$\text{Maximize } z = 1000x_1 + 1900x_2 + 2700x_3 + 3400x_4$$

(.....Next, write down all the constraints inequalities.....)

The optimum solution (obtained using TORA) is

$$\text{Total tax collection} = z = \$343,965$$

$$\text{Number of single homes} = x_1 = 35.83 \approx 36 \text{ units}$$

$$\text{Number of single homes} = x_2 = 98.53 \approx 99 \text{ units}$$

$$\text{Number of single homes} = x_3 = 44.79 \approx 45 \text{ units}$$

$$\text{number of quadruple homes} = x_4 = 0 \text{ units}$$

$$\text{number of homes demolished} = x_5 = 244.49 \approx 245 \text{ units}$$

Exercise 6.

1. (**Planning**) A company manufactures two products X and Y , which require the following resources. The resources are the capacities of machines M_1 , M_2 and M_3 . The available capacities are 50, 25 and 15 hours, respectively in the planning period. Product X requires 1 hour of machine M_2 and 1 hour of machine M_3 . Product Y requires 2 hours of machine M_1 and 2 hours of machine M_2 and 1 hour of machine M_3 . The profit contribution of products X and Y are \$5 and \$4, respectively. Formulate and solve the mathematical (LP) model that will maximize the profit. Use graphical method.

2. (**Production Allocation Problem**) A firm produces three products. These products are processed on three different machines. The time required to manufacture one unit of each of the three products and the daily capacity of the three machines are given in the table below.

Machine	Time per unit (minutes)			capacity (minutes/pay)
M_1	2	2	2	440
M_2	4	-	3	470
M_3	2	5	-	430

It is required to determine the daily number of units to be manufactured for each product. The profit per unit for products 1,2 and 3 are \$4, \$3 and \$6, respectively. It is assumed that all the amounts produced are consumed in the market. Formulate and solve a mathematical (L.P) model that will maximize the daily profit.

** End **